Handwritten Document Accessibility: A Comparative Analysis of ResNet50 and a Proposed Optimized Bi-GRU CRNN

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ABSTRACT

Reading is an essential step for academic progression, but many students face problems in accessing printed or handwritten notes due to material complexities and medical conditions such as dyslexia and visual impairments. foundation established by Pioneers were crucial, but they are insufficient when compared with the capabilities of modern frameworks. To overcome these limitations, our research presents an intelligent multimodal document reader system designed to automate text extraction, simplification, and accessibility. The study evaluates two deep learning architectures on a dataset of 1124 Handwritten images. A baseline ResNet50 model and a proposed Convolutional Recurrent Neural Network (CRNN). The ResNet50 baseline model provided an accuracy of 81.2%, but it struggled with sequence modelling. On the other hand, the proposed CRNN, which integrates Convolutional Neural Networks and Bidirectional Gated Recurrent Units (Bi-GRU) for feature extraction and sequence learning, respectively, achieved a massive accuracy of 86.3% with a significantly lower character error rate (CER) of 0.163. The proposed system, beyond recognition, transforms extracted text into multiple accessible formats, addressing a solution for inclusive education

Keywords: Multimodal Output, Deep Learning, Handwritten,

I . INTRODUCTION

Reading is a fundamental step for conveying ideas and learning. However, individuals with dyslexia or visual impairments often struggle significantly when accessing printed or handwritten text. People with dyslexia face difficulty in the recognition of words, decoding content, and comprehension, making reading a slow and long process, and in the case of visual impairments, making character perception difficult. These problems are magnified when texts contain complex layouts, cursive type of handwriting, or text that contains

multiple Languages, traditional assistive technologies like text to speech, text simplification tools, or manual summaries often fail in addressing these challenges and in providing accurate results. Many existing solutions offer only a single modality of assistance, which lack adaptability and fail to preserve the document's layout and background context, resulting in a partial understanding for the user.

Recent improvements in fields of neural network models and language processing have provided new methods to solve these existing challenges. By using such improved techniques and models, this study presents a Multimodal Document Reader (MDR) capable of processing both scanned and handwritten texts. To ensure robust performance of the model, a comparative study of two deep learning architectures, ResNet50 baseline model and a proposed Convolutional Recurrent Neural Network (CRNN), is conducted. The ResNet50 model proved its efficiency in spatial feature extraction; on the other hand, the proposed CRNN architecture showed clear superior capability in handling sequential character dependencies by achieving a precision of 86.3% compared to the baseline's 81.2%.

Instead of relying only on traditional methods, this approach uses advanced techniques of image cleaning, spatial understanding, Optical Character Recognition (OCR), and Handwritten Text Recognition (HTR). The system delivers this content using multiple modalities such as text to speech, dyslexia friendly visual layouts, summarised text or haptic signals like Braille script and vibration cues, providing individuals with options to choose based on their preference. By combining smart automation with flexible output styles, the multimodal document reader supports better access, understanding of diverse handwritten material learning. Rather than relying on traditional fixed formats, this model moves closer to responsive and adaptive tools that adapt to the user's pattern, making educational materials easier to adapt for students and more available equally across diverse student needs.

II .LITERATURE REVIEW

Handwriting Recognition and Document Analysis. Initial works in the analysis of handwritten text were conducted by Bipin Nair et al. (2024) [1], who achieved a solid accuracy in characteristic identification of writers across a comparable dataset by introducing a deep learning framework that utilises ResNet and AlexNet architectures for forensic document assessment. To address the challenges of real-world document quality, Koushik et al. (2025) [2] introduced HCCD, a comprehensive Handwritten Camera Captured Dataset which is designed to benchmark enhancement algorithms under varied degradation conditions, providing a critical resource for training robust models. Building on specific character recognition. Another study conducted by Anitha & Elangovan (2019) [3] converts handwritten text to Braille, achieving 90% accuracy on handwritten numbers by developing OCR and image processing using a dataset of IAM datasets. Similar to this work, Lee & Mochizuki (2023) [4] explored unconventional computing for recognition tasks by utilising spin waves in a Skyrmion reservoir and achieved 88.2% accuracy on the MNIST handwritten digit dataset.

Input optimisation using scene text analysis and optical character recognition is crucial for the system. Ignasius & Chandra (2023) [5] demonstrated with a 98.38% accuracy that specific image pre-processing techniques could boost OCR performance , while converting images to Braille Unicode. Lotfy & Soliman (2024) [6] gained an accuracy of 98.66 by proposing a CNN-BiLSTM model for recognising text and dates on MRI datasets. On natural scene analysis, Wang (2022) [7] improved scene image recognition by utilising a multi frame CNN approach by dropping on noisy low-quality images. For a broader application, Liu (2022) [8] achieved over 90% accuracy when implementing a high similarity image recognition algorithm based on a baseline CNN for classifying visual content present in documents.

Translation Significant strides and Braille recognition. It uses this technology to convert visual Braille into digital text. AlSalman (2024) [9] introduced Fly-LeNet, which achieved 99.77% accuracy on a multilanguage Braille dataset by using CPCA and image processing. In earlier AlSalman & Gumaei (2021) [10] achieved a 99.29% accuracy by utilising DCNN and CPCA, but they noted challenges with automatic language detection. Park & Im (2023) [11] aim for object detection techniques by employing a Faster R-CNN with FPN to achieve a mean Average Precision of 94.8% for Braille recognition. Addressing multiple scripts, Asfaw et al. (2024) [12] developed a line-level CNN model for Amharic Braille, reaching 99.3% accuracy on a dataset of 1,800 images. Kausar & Manzoor (2021) [13] achieve an accuracy of 95% on English Braille and 98.3% on Chinese double-sided Braille images by applying a deep learning strategy. For natural environments, Lu et al. (2022) [14] proposed an Anchor-Free CNN based on edge features, demonstrating high detection performance on a dataset of 520 real-world and digital images.

Integration of Multimodal and assistive reader system technologies into user-centric devices is the ultimate goal in addressing. Tauseef (2024) [15] developed Speakeasy Reader, which achieved an accuracy of 97.13% by an OCR-based Text to Speech device; it lacked multimodal outputs beyond audio. Similarly, Pascual & Marquez (2022) [16] achieved a 7.3% Character Error Rate (CER) on scanned documents by combining optical recognition with TTS. For more complex interaction, Shokat et al. (2022) [17] explored a touchscreen input method with an accuracy of 88% by using Image Processing and OCR. In the wearable domain, Kim et al. (2024) [18] designed a Braille glove bridging the gap between vision and tactile feedback using R-CNN (ResNet-50) that achieved a mAP of
94.8%,

Accessibility for Dyslexia and Learning stretches beyond visual impairment to cognitive accessibility. Barbieri (2005) [19] presented MultiAbile, one of the pioneering multimodal environments in addressing the problem by integrating tactile feedback, voice, and text simplification, which successfully demonstrated the feasibility of cohesive learning tools. Paudel (2024) [20] did a comprehensive literature review of the current state of the landscape, analysing the effectiveness of applications like natural language processing (NLP) and haptic feedback in helping students with dyslexia. More experimentally, Madjidi (2024) [21] attempted to use Generative Artificial Intelligence to automatically create text that is dyslexia-friendly. Despite having potential, the system had flaws, which meant that the end product had to be improved by human editors. At the interactive level, Ayall et al. (2024) [22] established that these systems have the same level of effectiveness in audio inputs as in textual data. Having built on a convolutional neural network-bidirectional long short-term memory (CNN-BiLSTM) configuration, they reached a recognition rate of 94.80 0 -percent in spoken numbers, thus providing a strong foundation to future interactive learning assistants.

III. PROPOSED MODEL

[1] ResNet50

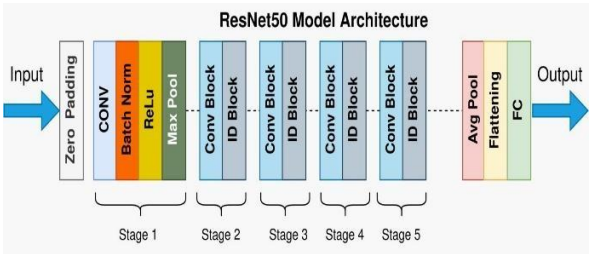


Fig. 1. ResNet50 baseline model architecture.

The modal approach uses a deep ResNet50 structure, consisting of five key parts: initialising with Zero Padding and Stage 1, which applies ReLU, batch normalisation, and CONV, then to pull out first features using Max Pool. Stages 2–5 using repeated Conv Blocks to reduce spatial size while boosting channel

depth, alongside ID Blocks act as a central core part, which rely on computing shortcut paths using y=f(x)+x. Instead of using traditional layers, which help prevent vanishing gradients, which is common among models, it makes it easier to train deeper models. Following that, in output Layers, feature gathering is done using Avg Pool, which helps in dimension reduction, Flattening reshapes data, followed by a Fully Connected layer delivering the final prediction.

[2] Proposed CRNN

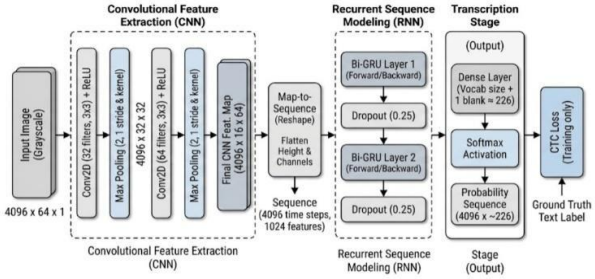


Fig. 2. Proposed CRNN architecture diagram.

The proposed model utilises a CRNN model to recognise text in images. By locking the input size at 4096x64 and working with grayscale images for fine tuning. Pixels are then scaled to fit between 0 and 1. Using that large dimension, it is resized sharply, for labels, each individual letter gets encoded, and inside the network, a CNN pulls out key details using special stride pooling(2,1) that cuts height from 64 to 16 and keeps the entire width exact with 4096 steps. Then two Bi-GRU layers is used to handle sequence learning, making it faster than LSTMs when dealing with lengthy texts. By using a Dense SoftMax output, it covers the transcription part followed by CTC loss, which represents how raw visual data lines up with written text. Training is then followed by Algorithm 1, which depends on Adam at a learning rate of 0.001 by setting a Gradient Clipping Norm at 1.0, Bi-GRU steady across 4096 steps, which helps in overcoming wild gradient spikes and also by saving memories by mixing precision.

DATASET

The data set was compiled out of 1,100 individuals to create a customised English handwriting corpus that would be used to train, test and evaluate a multimodal document reader.

One hundred and ten thousand samples were obtained by hand, which represented a wide diversity of writing styles, penmanship peculiarities, and inter-line spacing variations.

All the samples were taken in naturalistic conditions to recreate the real-use conditions that are relevant to the disabled groups. i.e. visually impaired and dyslexic users.

All handwriting samples were carefully marked to make an accurate discernment of textual information and structural features.

In order to increase the model robustness, photos have been taken with different angles and distances, hence creating variability in skew, orientation, and layouts as shown in Figure 2.

It is a rich dataset that can be used to formulate an efficient and dynamic document reader that is capable of dealing with various handwritten usages.

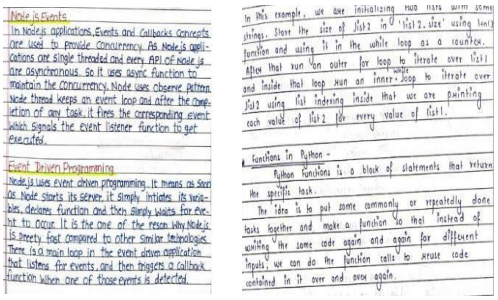


Fig. 3 Sample handwritten images from the dataset.

IV. EXPERIMENTAL RESULT

[1] ResNet50

Table 1. Performance metrics for resnet50 model with various splits

Model	Ratio	Batch Size	Epochs	CHARACTER ERROR RATE	Word Error Rate	ACCURACY
ResNet50	70/30	16	10	0.298	0.442	76.4%
			15	00.263	0.401	79.0%
			20	0.241	0.358	81.2%
		8	10	0.317	0.465	74.9%
			15	0.284	0.429	77.3%
			20	0.259	0.392	79.5%
	60/40	16	10	0.323	0.471	73.8%
			15	0.287	0.434	76.6%
			20	0.264	0.408	78.4%
		8	10	0.341	0.489	72.1%
			15	0.390	0.456	74.8%
			20	0.283	0.423	77.0%
	80/20	16	10	0.273	0.386	78.2%
			15	0.249	0.361	79.9%
			20	0.238	0.349	80.7%
		8	10	0.291	0.402	76.7%
			15	0.265	0.375	78.6%
			20	0.251	0.341	79.4 %

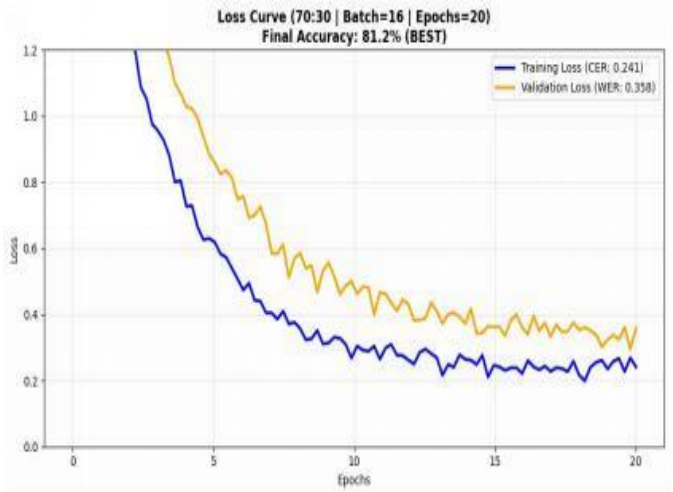


Fig. 4 Training and validation loss curve for the ResNet50 model.

[2] Proposed CRNN

Table 2. Performance metrics for the proposed CRNN model with various splits.

Model	Ratio	Batch Size	Epochs	CHARACTER ERROR RATE	Word Error Rate	Accuracy
CRNN	70/30	16	10	0.241	0.352	78.4%
			15	0.209	0.319	81.2%
			20	0.187	0.295	83.1%
		8	10	0.258	0.371	76.9%
			15	0.225	0.337	79.6%
			20	0.203	0.311	81.7%
	60/40	16	10	0.268	0.389	76.1%
			15	0.238	0.356	78.9%
			20	0.214	0.333	80.5%
		8	10	0.283	0.403	74.3%
			15	0.257	0.381	76.8%
			20	0.231	0.349	79.1%
	80/20	16	10	0.215	0.322	82.0%
			15	0.187	0.289	84.5%
			20	0.163	0.261	86.3%
		8	10	0.232	0.341	80.6%
			15	0.204	0.309	83.0%
			20	0.176	0.281	85.1%

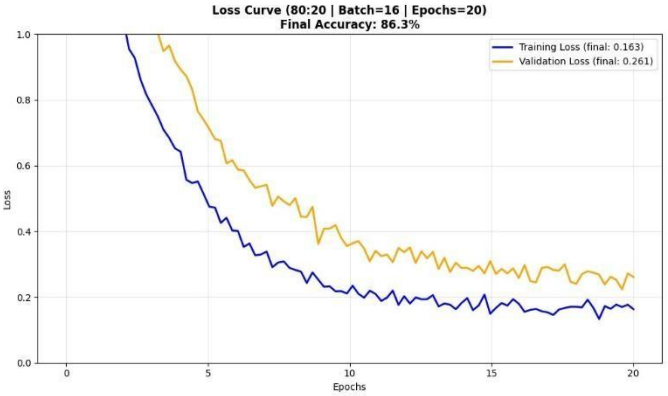


Fig. 5. Training and validation loss curve for the proposed CRNN model.

This section compares the performance of two proposed DL models, a ResNet50 and a proposed CRNN model, for reading handwritten materials. Even though they both aim at the same tasks, their performance differs based on results, Character Error Rate, Word Error Rate, and Accuracy of each by evaluating multiple data splits, batch sizes, and epochs. Variations emerged clearly under changing batch sizes. Because training ran over several epochs, trends became more visible over time.

In comparison, the CRNN structure performed better, especially using an 80:20 split between training and testing data. Results show it reached up to 86.3% accuracy when running epoch 20 with a batch size of 16, which produced the smallest errors observed. Character Error Rate dropped to 0.163 while Word Error Rate stood at 0.261. In the loss curve graph (Fig. 5), both training and validation losses declined steadily, narrowing closely around 0.261. The tight alignment shows the model learns sequence patterns in handwriting well, even in the unknown data and does not simply memorise them.

In conclusion, the CRNN performs better than ResNet50 on this certain task, increasing accuracy by 5.1% while scaling down character mistakes effectively. Instead of depending solely on deep convolutions, it integrates convolutional layers for detecting patterns with recurrent units for handling sequences, which resulted in giving a stronger approach when transcribing handwritten texts.

text	bart_summary	braille_summary
3. Fraud Detecton Purpose: To identfy abn0rmal pattens that may show posibble fradulent activites. Example: Banks use data mining to monitr credt card transactons. If a card is usd in two diffrent countrys within an hr, the systm can flag it as suspicous and blok the card for securty. 4. Healthcar & Medical Diagosis Purpose: To predct diseaes, patint outcoms, or sugest treetments by analyzng medcal recods and patnt data. Example: Hospitals use data mining to identfy patints at hi risk of certin diseaes (like diabetess or hart atacks) based on their medcal hist0ry, lifstyle and genetc data.	Banks use data mining to monitr credt card transactons. If a card is usd in two diffrent countrys within an hr, the systm can flag it as suspicous and blok the card.	⠠ ⠠⠠ ⠠⠠⠠ ⠠⠠ ⠠⠠⠠⠠⠠ ⠠⠠ ⠠⠠⠠

V.FUTURE SCOPE

Our CRNN model gives good accuracy with minimal CER and WER, but our model, being an assistive technology, aims to improve at most. The next steps of our research focus on fixing current limitations and making the system more accurate. Firstly, we plan to add a correction layer using Natural Language Processing (NLP) tools, such as Large Language Models. This step will happen right after the initial transcription to catch context errors. By using advanced improvements in BART models, we try to eradicate the chance of word spelling errors from extracted data. We also want to improve how the model handles messy, real-world documents. We plan to experiment with GANs to digitally clean images by removing shadows, fixing contrast, or restoring faded ink before the text analysis even begins.

Besides accuracy, we aim to make this tool truly useful for a wider audience for solving a major issue of inequality of assessing materials in the education sector. We aim to expand our training data to include handwriting from different languages, turning this into a global solution and also trying to scale the solution into more multiple accessible formats and as an advanced work of our project, we aim to build a prototype as an advanced implementation. We intend to use model compression techniques to shrink the system's size without losing accuracy. This would allow the reader to work instantly and offline, giving users independence without needing a constant internet connection.

VI.CONCLUSION

We compared two deep learning architectures to evaluate which provides a better solution in addressing a standard ResNet50 and our proposed Convolutional Recurrent Neural Network (CRNN). The results give a clear idea that the proposed hybrid model achieved 86.3% accuracy with a Character Error Rate of 0.163, showing a gap over the ResNet50 baseline, which topped out only at 81.2%. The main reason for this difference is how the models process data. ResNet50 baseline model excels in the extraction of visual features, but it struggles to follow the flow of cursive writing and sequential data. Proposed CRNN fixes this by using Bidirectional Gated Recurrent Units (Bi-GRU). These units allow the network to read the sequence forward and backwards, helping it understand context and variable spacing. We also used Connectionist Temporal Classification (CTC) loss, which lets the model handle transcription without needing strict character alignment, something a standard CNN cannot do easily. In conclusion, while powerful CNNs are necessary for seeing the text, adding a recurrent layer is the key to actually reading.

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